# UniFind

## Campus Lost & Found Website

### Business Analysis Requirements Document

| Field | Details |
|---|---|
| Project | UniFind – Campus Lost & Found Website |
| Document Type | Business Analysis Requirements Document |
| Role | Business Analyst |
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## 1. Project Overview

The UniFind – Campus Lost & Found Website is a web-based platform designed to help university students report lost and found items, search for relevant reports, upload item images, identify possible matches using image matching technology, and communicate with other users regarding potentially recovered items. The website also provides administrative functions for managing users, reports, flags, and disputes.

## 2. Problem Statement

Students often face difficulties when trying to recover lost belongings within the university environment. Traditional methods such as notices, social media posts, and word-of-mouth can make it difficult to identify matching items and communicate with the correct owner. The proposed website provides a centralized digital platform where students can report, search, match, and communicate about lost and found items.

## 3. Project Objectives

- Allow students to report lost items.
- Allow students to report found items.
- Provide an easy way to search for lost and found items.
- Allow users to upload item images.
- Identify possible matching items using image matching.
- Allow users to communicate about possible matches.
- Allow users to flag inappropriate content or behavior.
- Allow administrators to manage users, reports, flags, and disputes.
- Provide a reliable, responsive, and user-friendly campus lost-and-found website.

## 4. Stakeholders

| Stakeholder | Role / Interest |
|---|---|
| **Students** | Main users who report, search, match, and recover lost or found items. |
| **System Administrator** | Manages users, reports, flags, disputes, and administrative actions. |
| **Project Team** | Analyzes, designs, develops, tests, and maintains the website. |
| **University** | Provides the campus environment, policies, and governance context. |
| **Business Analyst** | Elicits, analyzes, documents, prioritizes, and validates requirements. |
| **QA / Tester** | Verifies that implemented website functionality satisfies acceptance criteria. |

## 5. System Actors

| Actor | Responsibilities |
|---|---|
| **Student / User** | Register, log in, manage profile, report items, upload images, search, view matches, communicate, flag content, submit disputes, and provide feedback. |

| | |
|---|---|
| **Administrator** | Manage users and reports, review flags and disputes, and take appropriate administrative action. |
| **Image Matching System** | Analyze and compare item images and return possible matches with similarity information. |
| **Notification System** | Deliver relevant notifications such as possible matches, messages, and status updates. |
| **Web Browser** | Provides the interface through which users access the web-based website. |

# 6. Project Scope

## 6.1 In Scope

- User registration and authentication
- User profile management
- Lost item reporting
- Found item reporting
- Item image upload and validation
- Item search and filtering
- Item details and status management
- Image-based matching
- Possible match results
- Communication between authorized users
- Notifications
- Flagging users or reports
- Dispute submission and resolution
- Reviews / feedback
- Administrative management

## 6.2 Out of Scope

- Physical delivery of recovered items
- Buying or selling items
- Payment processing
- External marketplace integration
- Advanced facial recognition
- Automatic physical verification of ownership

## 6.3 Platform and Access

The system will be developed as a web-based website. Users will access the website through a modern web browser using an internet connection. The website will provide responsive and user-friendly interfaces suitable for desktop and mobile screen sizes.

## 7. Functional Requirements

| ID | Functional Requirement | Priority |
|---|---|---|
| **FR-01** | The system shall allow students to register an account. | Must |
| **FR-02** | The system shall allow registered users to log in and log out. | Must |
| **FR-03** | The system shall allow users to manage their profiles. | Should |
| **FR-04** | The system shall allow users to report a lost item. | Must |
| **FR-05** | The system shall allow users to report a found item. | Must |
| **FR-06** | The system shall allow users to upload images of lost/found items. | Must |
| **FR-07** | The system shall allow users to edit and delete their own item reports. | Must |
| **FR-08** | The system shall display lost and found item listings. | Must |
| **FR-09** | The system shall allow users to search for items by keywords. | Must |
| **FR-10** | The system shall allow users to filter items by category, location and date. | Should |
| **FR-11** | The system shall compare uploaded item images with existing item images. | Must |
| **FR-12** | The system shall display possible image matches ranked by similarity. | Must |
| **FR-13** | The system shall allow users to view detailed information about an item. | Must |
| **FR-14** | The system shall allow users to contact another user regarding a possible match. | Should |
| **FR-15** | The system shall allow users to send and receive messages. | Should |
| **FR-16** | The system shall notify users about relevant updates or possible matches. | Should |
| **FR-17** | The system shall allow users to flag inappropriate reports or users. | Should |
| **FR-18** | The system shall allow users to submit disputes. | Should |
| **FR-19** | The system shall allow users to provide reviews/feedback after an interaction. | Could |
| **FR-20** | The system shall allow administrators to manage users. | Must |
| **FR-21** | The system shall allow administrators to review flagged content/accounts. | Must |
| **FR-22** | The system shall allow administrators to review and resolve disputes. | Must |
| **FR-23** | The system shall allow administrators to warn, suspend or deactivate accounts. | Must |
| **FR-24** | The system shall maintain item statuses such as Lost, Found, Matched, Returned and Closed. | Must |

## 8. Non-Functional Requirements

| ID | Non-Functional Requirement | Priority |
|---|---|---|
| **NFR-01** | Normal user requests should receive a response within approximately 2–3 seconds under normal conditions. | Should |
| **NFR-02** | Image matching should provide results within an acceptable time. | Should |
| **NFR-03** | User passwords must be securely hashed and never stored as plain text. | Must |
| **NFR-04** | Protected features must require authentication. | Must |
| **NFR-05** | Users must only be able to modify their own reports. | Must |
| **NFR-06** | Private user information and conversations must only be accessible to authorized users. | Must |
| **NFR-07** | The system should minimize unexpected failures and data loss. | Must |

| | | |
|---|---|---|
| **NFR-08** | The website interface should be simple, responsive, consistent, and easy for students to navigate on desktop and mobile devices. | Must |
| **NFR-09** | The website should work correctly on supported modern web browsers such as Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari. | Should |
| **NFR-10** | The system should maintain accurate and consistent data. | Must |
| **NFR-11** | Important administrative actions should be recorded. | Should |
| **NFR-12** | Uploaded images should be validated for supported format and size. | Must |
| **NFR-13** | The system should display clear error messages when invalid information is entered. | Must |
| **NFR-14** | The system should be maintainable and structured for future development. | Should |

## 9. Business Rules

| Rule ID | Business Rule |
|---|---|
| **BR-01** | Only registered users can submit lost or found item reports. |
| **BR-02** | A user can edit or delete only their own reports. |
| **BR-03** | An item report must contain all required information before submission. |
| **BR-04** | An item may have a status of Lost, Found, Possible Match, Matched, Returned, or Closed. |
| **BR-05** | A user must not be able to access another user's private conversations. |
| **BR-06** | Administrators can review reports that have been flagged. |
| **BR-07** | Administrators can take action against accounts that violate system rules. |

## 10. Flagging Rules

A user or item report may be flagged when:

- A user submits false information.
- A user repeatedly creates misleading reports.
- A user uploads inappropriate content.
- A user harasses another user.
- A user behaves abusively.
- A user attempts fraudulent activity.
- A user violates applicable university rules.

Flagging Process:

1. User submits flag
2. System records flag
3. Administrator reviews flag
4. Administrator investigates

5. Administrator makes a decision
6. Action is taken if required
7. Flag is resolved

## 11. Administrative Actions

- Review flagged users.
- Review flagged reports.
- Review disputes.
- Send warnings.
- Temporarily suspend accounts.
- Permanently deactivate accounts.
- Remove inappropriate reports.
- Resolve disputes.

## 12. Dispute Management

A user may submit a dispute when:

- They believe an ownership claim is incorrect.
- An item was not returned as expected.
- Another user provided false information.
- There was inappropriate behavior.
- There is disagreement regarding a reported item.

Dispute lifecycle:

| Status | Meaning |
|---|---|
| **Open** | Dispute has been submitted and is awaiting review. |
| **Under Review** | Administrator is investigating the dispute and evidence. |
| **Resolved** | Administrator has completed the review and recorded a decision. |

Possible outcomes:

- No Action
- Warning
- Account Suspension
- Account Deactivation
- Report Removed

## 13. User Story Backlog Summary

| ID | Epic | User Story | Priority |
|---|---|---|---|
| **US-01** | User Management | Register Account | Must |
| **US-02** | User Management | Login | Must |
| **US-03** | User Management | Logout | Must |
| **US-04** | User Management | Manage Profile | Should |
| **US-05** | Lost & Found Item Management | Report Lost Item | Must |
| **US-06** | Lost & Found Item Management | Report Found Item | Must |
| **US-07** | Lost & Found Item Management | Upload Item Image | Must |
| **US-08** | Lost & Found Item Management | View Item Details | Must |
| **US-09** | Lost & Found Item Management | Edit Item Report | Must |
| **US-10** | Lost & Found Item Management | Delete Item Report | Must |
| **US-11** | Lost & Found Item Management | Update Item Status | Must |
| **US-12** | Image Matching | Upload Image for Matching | Must |
| **US-13** | Image Matching | Compare Item Images | Must |
| **US-14** | Image Matching | Display Possible Matches | Must |
| **US-15** | Image Matching | View Match Details | Must |
| **US-16** | Search & Discovery | Search Items | Must |
| **US-17** | Search & Discovery | Filter Items | Should |
| **US-18** | Search & Discovery | View Search Results | Must |
| **US-19** | Communication & Notifications | Contact User | Should |
| **US-20** | Communication & Notifications | Send Message | Should |
| **US-21** | Communication & Notifications | Receive Message | Should |
| **US-22** | Communication & Notifications | Receive Notifications | Should |
| **US-23** | Reviews & Feedback | Submit Feedback | Could |
| **US-24** | Reviews & Feedback | View Feedback | Could |
| **US-25** | Administration & Disputes | Manage Users | Must |
| **US-26** | Administration & Disputes | Review Item Reports | Must |
| **US-27** | Administration & Disputes | Review Flagged Accounts | Must |
| **US-28** | Administration & Disputes | Flag User or Report | Should |
| **US-29** | Administration & Disputes | Submit Dispute | Should |

| | | | |
|---|---|---|---|
| **US-30** | Administration & Disputes | Review Dispute | Must |
| **US-31** | Administration & Disputes | Resolve Dispute | Must |
| **US-32** | Administration & Disputes | Suspend User | Must |
| **US-33** | Administration & Disputes | Deactivate User | Must |

## 14. Sprint 1 Candidate Stories

The following stories are recommended as Sprint 1 candidates. Final Sprint 1 scope should be agreed with the development team and project stakeholders.

- Register Account
- Login
- Report Lost Item
- Report Found Item
- Upload Item Image
- View Item Listings
- Search Items
- View Item Details
- Basic Image Matching

## 15. Requirement Traceability

| Requirement(s) | Epic | Representative Stories | Business Area |
|---|---|---|---|
| **FR-01 / FR-02** | User Management | Register Account / Login | Account access |
| **FR-04 / FR-05** | Lost & Found Item Management | Report Lost Item / Report Found Item | Item reporting |
| **FR-06 / FR-11 / FR-12** | Image Matching | Upload / Compare / Display Matches | Image matching |
| **FR-09 / FR-10** | Search & Discovery | Search / Filter Items | Discovery |
| **FR-14 / FR-15 / FR-16** | Communication & Notifications | Contact / Message / Notifications | Communication |
| **FR-17 / FR-18** | Administration & Disputes | Flag / Submit Dispute | Governance |
| **FR-20–FR-24** | Administration & Disputes | Manage / Review / Resolve | Administration |

## 16. Assumptions and Constraints

### 16.1 Assumptions

- The primary users are university students.
- Users have access to a modern web browser and internet connection.
- Users provide reasonably accurate item information.

- The image matching functionality can be implemented using an appropriate technical approach.
- Administrative users are authorized by the university/project context.

### 16.2 Constraints

- Limited project time and resources.
- Image quality and the selected image-matching approach may affect matching accuracy.
- University privacy, security, and usage policies must be considered.
- The project focuses on core campus lost-and-found functionality rather than a general marketplace.
- The website must remain usable across supported modern browsers and common screen sizes.

Requirements should be reviewed and agreed before development. Changes after approval should be documented, assessed for impact, and communicated to the project team.

## 17. Requirements Approval

The requirements should be reviewed and agreed by the project team before implementation.

| Role | Responsibility | Name | Approval / Status | Date |
|---|---|---|---|---|
| **Project Manager** | Reviews and approves project scope and requirements | P. D. M. Wanigasekara | Approved | 07 October 2026 |
| **Business Analyst** | Prepares, documents, and maintains project requirements | M. A. P. S. Perera | Approved | 06 October 2026 |
| **QA Engineer** | Reviews requirements for clarity, completeness, and testability | K. M. D. Tishani | Approved | 07 October 2026 |
| **Developer 1** | Reviews requirements for technical feasibility | W. D. Sampath | Approved | 07 October 2026 |
| **Developer 2** | Reviews requirements for technical feasibility | H. D. M. Rukshani | Approved | 07 October 2026 |
| **Developer 3** | Reviews requirements for technical feasibility | W. R. A. P. Sakura | Approved | 07 October 2026 |